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Intellisample technology

While most of the pixel and vertex specifications are focused on DirectX 9, the pixie dust here is a technology called Intellisample that will make even games such as *DOOM III* run faster. Intellisample is a comprehensive set of technologies that includes a new colour compression engine, improved fast z-clear, dynamic gamma correction, adaptive trilinear and anisotropic filtering, and anti-aliasing.

The GeForce FX maximises its memory bandwidth by compressing all the data that comes out of the rendering pipeline before sending it to the memory controller (as described above). This results in a direct increase in effective memory bandwidth, allowing for larger and more complex textures. This is noticed especially when anti-aliasing is enabled, where the demands placed on the memory bandwidth are greater. There is a newer algorithm for clearing the z-buffer (for getting rid of obstructed or invisible polygons), resulting in faster frame processing. Finally, the card uses various methods for implementing filtering options. The user can choose between a direct filtering



The level of detail in the environment and skies is made realistic by the GeForce FX' ability to process a large number of polygons at high speeds

option (trilinear or anisotropic), or a less accurate but a high performance option. This will result in a lower performance hit compared to running the card in any one of these filtering modes directly.

Faster core and memory speeds

When it debuts, the GeForce FX is expected to have a core speed of 500 MHz and a memory speed of 250 MHz (effectively 1 GHz due to the 4x increase, being DDR2). This is significantly higher than the Radeon 9700's core speed of 325 MHz and 310 MHz DDR memory.

In newer games such as *Stalker* or *Ral-*



The enormous air intake duct and heat pipe solution help keep the GeForce FX cool

liSport, the realism in the models and environment is unprecedented due to the use of very high polygon-count. Also, the 128-bit colour support allows for hitherto unseen levels of accuracy in colours and specular highlights in the game.

FX flow

Given the high core and memory speeds, the card needs to breathe well. Hence, it uses an advanced cooling system involving the use of heat pipes. In addition to a heat sink on the critical heat-generating components such as the GPU and memory chips, there are tiny copper pipes that draw the heat away from these elements. This is implemented through a special air-flow duct in conjunction with the cooling fan, resulting in a large cooling assembly that takes up the space of two slots in your cabinet!

The last word

Every once in a graphics era comes a product that offers not only high clock speeds or more memory, but truly intelligent technology with a specific goal and uses innovative methods to attain it. nVidia's GeForce FX is one such product that represents a new species of graphics hardware built to redefine desktop graphics.

While all this technology scientifically translates into more pixels per second, greater colour depths and better filtering, the truth will be bared when we see shimmers on water and winking characters on our desktops. Though this card will be out of our financial reach, in time, the technology will trickle down to more affordable solutions. So be prepared for the time when you can feel the knot in the pit of your stomach while you watch a gleaming Lamborghini Murciélago tear down your screen—the only difference being that instead of watching this with a bag of popcorn, you'll be holding a joystick in your hand! 

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Old vs New

Specification	nVidia GeForce FX	ATi Radeon 9700 PRO	nVidia GeForce4 Ti4600
Chip technology	256-bit	256-bit	256-bit
Process	0.13 micron	0.15 micron	0.15 micron
Transistors	125 million	107 million	63 million
Memory bus	128-bit DDR2	256-bit DDR	128-bit DDR
Pure memory bandwidth	16 GBps	19.8 GBps	10.4 GBps
Pixel fillrate	4 Gigapixel/s	2.6 Gigapixel/s	1.24 Gigapixel/s
Anti-aliased Fillrate	16 billion AA samples/s	15.6 billion AA samples/s	4.8 billion AA samples/s
Max FSAA mode	8x	6x	4x
Triangle transform rate	350 milion triangles/s	325 M milion triangles/s	69 M milion triangles/s
AGP bus	1x/2x/4x/8x	1x/2x/4x/8x	1x/2x/4x
Memory	128/256 MB	128/256 MB	128 MB
GPU clock	500 MHz	325 MHz	300 MHz
Memory clock	250 MHz (1000 DDR2)	310 MHz (620 DDR)	325 MHz (650 DDR)
Memory	BGA 2.0 ns	BGA 2.9 ns	BGA 2.8 ns
Vertex shader	FP Array	4	2
Pixel pipelines	8	8	4
Texture units per pipe	1	1	2
Textures per texture unit	16	8	4
DirectX generation	9.0 (+)	9	8
Optimisations	LMA II optimised colour compression	Hyper Z III	LMA II